MACHINE LEARNING

Algorithms, Interactive Playgrounds & Clickable Links

Corrected edition. Website links below are embedded hyperlinks. If your PDF viewer does not open a link, copy the website address into Chrome.

1. Interactive playgrounds — start here

Website	What you can explore	Link
ML Playground	46 lessons and 48+ interactive demos: foundations, classical ML, deep learning and LLMs.	Open
Interactive ML	Editable 2D data; linear/logistic regression, trees, bagging, forests, boosting, neural nets and toy transformer.	Open
ML Playgrounds	Visual demos for linear regression, Gaussian processes, SVM, PCA and Bayesian linear regression.	Open
ML Visualization	Interactive explainers for regression, classification, ensembles, clustering, dimensionality reduction and neural networks.	Open
ML Hive Playground	Browser-based ML playground collection, including regression, clustering and reinforcement learning.	Open
ML / AI Arcade	Interactive studies: K-Means, DBSCAN, hierarchical clustering, GMM, SVM, KNN, logistic regression, PCA, t-SNE and optimizers.	Open
TensorFlow Playground	Change neural-network layers, neurons, activation and learning rate; watch training and decision boundaries.	Open
Amazon MLU-Explain	Visual explanations of neural networks, random forests, decision trees and bias-variance.	Open
Google ML Crash Course exercises	Interactive exercises for linear regression, gradient descent, logistic regression, classification metrics and data preparation.	Open
Seeing Theory	Interactive probability and statistics visualizations for ML foundations.	Open

2. Algorithm-wise practical guide

Regression

Linear Regression: Fit a line to editable points; inspect slope, intercept and loss. Links: Resource 1 | Resource 2 | Resource 3

Multiple / Polynomial Regression: Explore multiple features or curved fits; compare complexity. Links: Resource 1 | Resource 2

Ridge / Lasso / Elastic Net: Understand L1/L2 regularization and coefficient shrinkage. Links: Resource 1 | Resource 2

Decision Tree Regressor: Watch feature splits partition the feature space. Links: Resource 1 | Resource 2

Random Forest / Extra Trees Regressor: Compare many trees and averaged predictions. Links: Resource 1 | Resource 2

Gradient Boosting / AdaBoost Regressor: See sequential learners correct earlier errors. Links: Resource 1 | Resource 2

SVR: Study support vectors and the epsilon-insensitive tube. Links: Resource 1

KNN Regressor: Change K and see how nearby samples influence predictions. Links: Resource 1 | Resource 2

Classification

Logistic Regression: Inspect sigmoid probabilities and move points to change the boundary. Links: Resource 1 | Resource 2

KNN Classifier: Change K and observe class regions around data points. Links: Resource 1 | Resource 2

SVM / SVC: Explore separating boundaries, margins and kernels. Links: Resource 1 | Resource 2

Decision Tree Classifier: Follow feature splits; change depth and observe the boundary. Links: Resource 1 | Resource 2

Random Forest / Extra Trees Classifier: Compare a single tree with an ensemble vote. Links: Resource 1 | Resource 2

Naive Bayes: Understand class probabilities and feature likelihoods. Links: Resource 1 | Resource 2

Gradient Boosting / AdaBoost: Explore ensemble learning and error correction. Links: Resource 1 | Resource 2

MLP Classifier: Experiment with neural-network decision boundaries. Links: Resource 1 | Resource 2

Clustering

K-Means: Move points and centroids; change K and watch clusters form. Links: Resource 1 | Resource 2

DBSCAN: Change epsilon and minimum samples; identify clusters and noise. Links: Resource 1

Agglomerative / Hierarchical: Explore how clusters merge into a hierarchy. Links: Resource 1 | Resource 2

Gaussian Mixture Models: Understand probabilistic clusters and soft membership. Links: Resource 1 | Resource 2

Mean Shift / OPTICS / BIRCH: Compare density-based and hierarchical clustering. Links: Resource 1

Dimensionality reduction

PCA: Project data onto principal components; inspect retained variance. Links: Resource 1 | Resource 2 | Resource 3

t-SNE / MDS / Isomap: Explore low-dimensional embeddings and neighborhood structure. Links: Resource 1 | Resource 2

Truncated SVD / NMF / ICA: Study matrix factorization and component extraction. Links: Resource 1

Autoencoders: Explore compressed representations and reconstruction. Links: Resource 1

Neural networks, optimization & reinforcement learning

Neural networks / activations / backpropagation: Change layers, neurons, activation and learning rate. Links: Resource 1 | Resource 2

Gradient Descent / SGD / Momentum / RMSProp / Adam: Watch parameter updates and loss change. Links: Resource 1 | Resource 2

Q-Learning / Reinforcement Learning: Experiment with rewards, actions and learned policies. Links: Resource 1

3. Official scikit-learn documentation

All estimator classes and functions: API reference

User guide: Scikit-learn user guide

Linear models: Linear models | Trees: Decision trees | SVM: SVM

Clustering: Clustering | Model selection: Cross-validation and tuning

4. Python functions to practice

Preprocessing: StandardScaler(), MinMaxScaler(), OneHotEncoder(), LabelEncoder(), SimpleImputer(), PolynomialFeatures().

Splitting/training: train_test_split(), model.fit(X_train, y_train), model.predict(X_test), model.score().

Regression metrics: mean_absolute_error(), mean_squared_error(), root_mean_squared_error(), r2_score().

Classification metrics: accuracy_score(), precision_score(), recall_score(), f1_score(), confusion_matrix(), classification_report(), roc_auc_score().

Validation/tuning: cross_val_score(), KFold(), StratifiedKFold(), GridSearchCV(), RandomizedSearchCV().

5. Suggested learning order

Linear Regression → Logistic Regression → KNN → SVM → Decision Trees → Random Forest → Naive Bayes → Gradient Boosting → K-Means → DBSCAN → PCA → Neural Networks → Gradient Descent/optimizers → Reinforcement Learning.

Learning rule: For each algorithm, visualize it first, implement it in Python second, and explain what changes when you alter its parameters.